

Afsan Sarwer

DVM, MSc Researcher | Microbial Genomics | Antimicrobial Resistance | Computational Biology

📍 Mymensingh, Bangladesh

Education

MSc in Bangladesh Agricultural University , Microbiology	Mymensingh, Bangladesh
Micro- • Department of Microbiology and Hygiene	Jan 2024 – Oct 2026
biol- • MTR Lab, supervised by Professor Dr. Md. Tanvir Rahman	
ogy • Expected completion: October 2026	
Doc- Bangladesh Agricultural University , Veterinary Science	Mymensingh, Bangladesh
tor of • CGPA: 3.314/4.00	Jan 2018 – Jan 2024
Vet-	
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nary	
Med-	
icine	
(DVM)	

Research Experience

Bangladesh Agricultural University , Comparative Genomics and Genomic Epidemiology of Escherichia coli	Mymensingh, Bangladesh
• Comparative whole-genome analysis of Escherichia coli isolates from vegetables and seafood.	Jan 2024 – present
• Analysis of antimicrobial resistance genes, virulence genes, plasmids, mobile genetic elements, MLST, serotypes, and genomic relationships.	
• Pangenome, phylogenomic, and genomic epidemiology analyses using bacterial whole-genome sequencing data.	
• Plasmid profiling and similarity analysis to investigate antimicrobial resistance dissemination.	
Bangladesh Agricultural University , Machine Vision and Deep Learning for Veterinary Parasitology	Mymensingh, Bangladesh
• Developed a computer vision approach for detection of livestock parasite eggs and cysts in microscopic images.	Jan 2023 – Jan 2024
• Implemented YOLOv8 using Python and evaluated model performance using standard classification and object-detection metrics.	

Publications

Genome-wide analysis of extended-spectrum beta-lactamase-producing Escherichia coli from seafood in Bangladesh: population structure, resistome, virulome, and global dissemination patterns	Jan 2026
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Conference Presentations

Genomic Epidemiology and Plasmid-Associated Antimicrobial Resistance of Vegetable-Derived Escherichia coli: A Nationwide Comparative Analysis	Bangladesh University,	Agricultural Mymensingh, Bangladesh
• Poster presentation, BSVER ASCon XXXIV.		Aug 2026
• Authors: Afsan Sarwer, Naeem Ahammed Ibrahim Fahim, Md. Liton Rana, Amrita Pondit, Saifur Rahman, Ricardo Oliva, and Md. Tanvir Rahman.		

Whole-Genome-Based Global Phylogeny, Population Structure, and Resistome Profiling of ESBL-producing *Escherichia coli* from Seafood in Bangladesh

Gazipur, Bangladesh
Feb 2026

- Poster presentation, Gazipur Agricultural University International Conference.
- Authors: Naeem Ahammed Ibrahim Fahim, Afsan Sarwer, Zannatul Firdous, Md. Saiful Islam, Saifur Rahman, Jayedul Hassan, Timothy R. Walsh, Refath Farzana, and Md. Tanvir Rahman.

Whole-Genome-Based Global Phylogeny, Population Structure, and Resistome Profiling of ESBL-producing *Escherichia coli* from Seafood in Bangladesh

Shahjalal University of Science and Technology, Sylhet, Bangladesh
Oct 2025

- Poster presentation, International Conference on Life Science.
- Authors: Naeem Ahammed Ibrahim Fahim, Afsan Sarwer, Zannatul Firdous, Md. Saiful Islam, Saifur Rahman, Jayedul Hassan, Timothy R. Walsh, Refath Farzana, and Md. Tanvir Rahman.

Livestock Endoparasite Detection with Machine Vision and Deep Learning

National Science and Technology Complex, Agargaon, Dhaka, Bangladesh
Nov 2024

- Poster presentation.
- Authors: Afsan Sarwer and Md. Fahim Shahria.

Technical Skills

Genomics and Bioinformatics: Whole-genome sequencing analysis; comparative genomics and pangenomics; phylogenetics and genomic epidemiology; plasmid and mobile genetic element analysis; antimicrobial resistance and virulence profiling.

Computational: Python; R; Linux and Bash; statistical analysis; data visualization; machine learning and deep learning.

Software and Tools: Anvi'o; PanACoTA; Roary; OrthoFinder; Parsnp; kSNP4; Snippy; Gubbins; IQ-TREE; RAxML; MOB-suite; Cytoscape; seqkit; Biopython.